



Viscosity of Fluids

Group #1

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Introduction

- The objective of this experiment was to model the viscosity of a CARBOPOL and water solution.
- Using different pressures and capillary tube sizes, the goals were to determine the viscosity of the solution and to model the viscosity versus shear rate. The Ostwald deWaele equation will be used.



Theory

Viscosity is a fluid property that measure a fluids ability to resist shear force

Newtonian

- Viscosity is constant
- Linear relation between shear stress and shear rate

$$\tau_{yx} = -\mu \frac{dv_x}{dy}$$

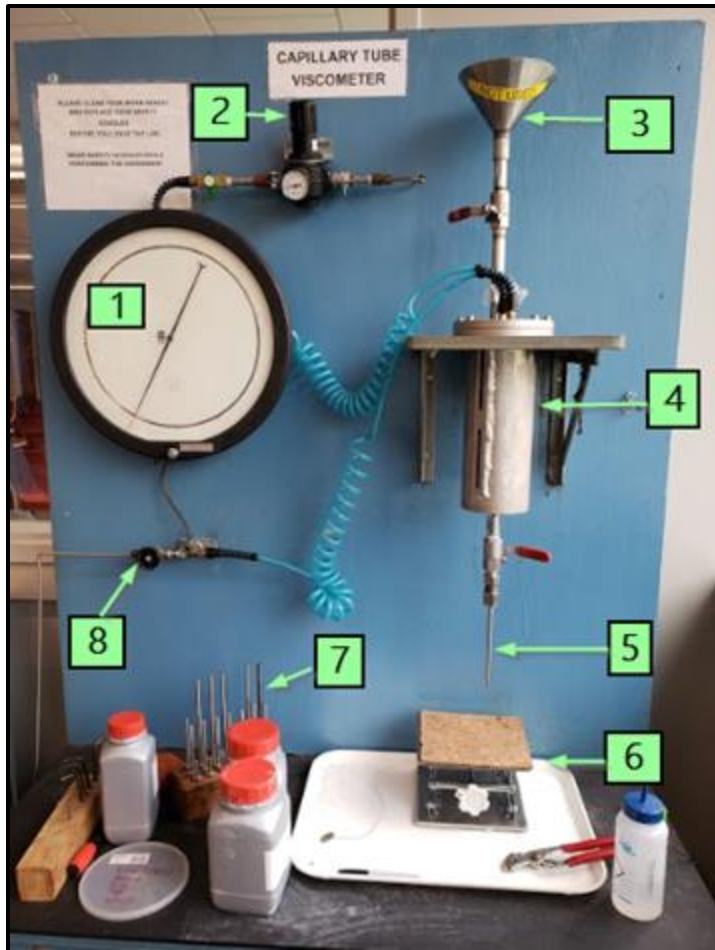


Non-Newtonian

- Apparent viscosity depends on shear rate
- Nonlinear relation between shear stress and shear rate
- Ostwald- deWaele model uses empirical constants m & n

$$\tau_{yx} = -m \left| \frac{dv_x}{dy} \right|^{n-1} \frac{dv_x}{dy} = -\mu_{app} \frac{dv_x}{dy}$$

Apparatus



Component #	Function
1 - Pressure meter	Shows how much pressure is present in the vessel.
2 - Pressure changer	When turned will increase or decrease pressure in vessel .
3 - Funnel and valve	Where the fluid is added to get to the vessel . It is not used
4 -Vessel	Contains fluid for which the viscosity is to be measured. It is pressurized with air
5 -Outlet valve	When opened , it will release the fluid from the vessel.
6 - Standing plate	Where we place the jar to collect the fluid leaving the vessel
7 - Capillary tubes (of different length : diameter ratio)	Where non-newtonian fluid passes, each will give different flow rates
8 - Vent valve	It is opened to decrease pressure in vessel, closed to increase pressure in vessel

Material & Supplies



- **Warm Tap Water:** to mix with CARBOPOL powder
- **CARBOPOL:** 8 wt% used for experiment
- **NaOH:** to neutralize solution to a pH between 5 and 6
- **Power Mixer:** to mix CARBOPOL and warm water
- **Stainless steel bucket:** mixing vessel for solution
- **Nitrile Gloves:** needed when working with the chemicals
- **Lab Coat optional:** Can wear to minimize mess
- **Balance:** used for weighing
- **Air:** used to pressurize the system
- **Pincers:** used to tighten or loosen capillary tubes from holding
- **Paper towels:** to clean
- **Beakers or cylinders:** for measuring volume of solution during experiment
- **Speed controller:** used to vary the mixing speed when making solution

Procedure



1. With Mixture created during your pre-lab date, Add to vessel.
2. Tighten Nuts for Vessel, Make sure all valves are closed.
3. (5) in the system is where you will change all capillaries sizes. Once a capillary is attached; adjust pressure valve.
4. Once the pressure is to the desired pressure open the valve that is in between pressure meter and pressure changer.
5. Once the pressure is to desired pressure open outlet valve. Collect fluid for 20 seconds in graduated cylinder then close valve and record mass of dispensed.
6. Close Pressure valves then repeat steps 2-5.

Experimental Challenges



Experimental Challenges / Lessons Learned

- Make solution during prelab
 - Use mixer for water and CARBOPOL, not for mixing base into solution
- Make sure scale is set to grams
- Try not to use the capillary tubes with the smallest diameters
- Use pressure >20 psig, a good starting point is 30 psig
- Close vent valve before pressurizing the system
- As you increase pressure per capillary tube, you don't have to depressurize every time, only before using the next capillary tube
- Calculate density during lab

Safety



- Wear safety glasses
- Wear gloves
- Depressurize system before changing capillary tubes
- Beware of hazards in making the solution as it is initially acidic and NaOH is added

Comment Summary

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2. 97/100

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3. Put some technical details
4. 13/15

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5. Show the final equation used to obtain m and n
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